Authentication and authorisation

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

Ask who established the role and which operation the policy permits.

Use denied-before-body traces and separate permission from actual execution; distinguish consolidation from fresh transfer.

Keep one habit: Model authority-source reasoning → inspect local policy traces → collect individual explanations → assess a fresh action/resource policy later.

MAIN LESSON · SECTION 1

Prepare the connected service investigation

> Keep modelling, shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Explain the **mechanism** and terminology of **authentication** and **authorisation**

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Opening retrieval

Without opening the old lesson, explain one key idea from Configuration and secrets. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of configuration and secrets

Reference example, not the only acceptable wording: Teacher explanation: Configuration supplies non-secret settings; secrets provide privileged values. Keep secrets out of source, logs and client responses.

The toy service reads DATA_VERSION from configuration and a signing secret from server-only **input**. Validate required settings at startup and fail clearly if absent; do not substitute a published default secret.

Without opening the old lesson, explain one key idea from Architecture and data flow. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of architecture and data flow

Reference example, not the only acceptable wording: Teacher explanation: Architecture assigns responsibilities to components and shows information crossing trust boundaries. A diagram should include failure paths.

Trace request validator -> role filter -> source lookup -> response validator -> result. Sources enter as data, not control instructions. Reject invalid requests before any downstream lookup.

Pictured end goal

Explain **authentication** and **authorisation** using the deciding step in this fictional service.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Configuration and secrets

Architecture and data flow

Use the three-session teaching cycle above, with extra time as needed. **Authentication** identifies an actor; **authorisation** decides whether that actor may perform a particular operation on a particular resource. Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions. Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

Outcomes

Explain the **mechanism** and terminology of **authentication** and **authorisation**

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach trusted identity context and action/resource permission separately

> Do not equate a local policy helper with deployed authentication.

Definitions and complete school read policy

Authentication identifies an actor; **authorisation** decides whether that actor may perform a particular operation on a particular resource.

Use trusted session roles, never a role claimed inside user JSON. Student may read student resources; teacher may read both. Unauthenticated requests are denied.

Question	School-service policy used here
Who is the actor?	Read the trusted session fixture; client-written role is not authoritative
What is requested?	Read one declared resource
Student role	May read student resources; may not read teacher-only resources
Teacher role	May read student and teacher resources
No recognized trusted session or unknown role	Deny under this service policy
Unknown resource classification	Deny rather than guessing permission
Permission decision	Must precede reading the resource body
Scope	Local policy exercise; no real login, cookies, expiry or writes

Main authority-source trace and checked reasoning

The trusted role wins over the role written in the request

STEP	SUPPLIED EVIDENCE AND DECISION
1. Inspect trusted context	Server-side classroom session says student
2. Inspect requested action/resource	Request asks to read a teacher-only guide and includes role=teacher
3. Choose the authority source	Use trusted session role; client-written role does not promote it
4. Apply read policy	Student cannot read teacher resource → deny before body lookup
5. Record the outcome	No guide body returned; trusted session remains student

The declared role is a fixture in this exercise. No password or identity-verification protocol is demonstrated.

Require learners to name the trusted session as the authority source and the request role as untrusted data. The simulated role fixture is supplied; no identity verification has actually run. Inspect denied traces for absence of body reads. For the museum, an allowed authorization gate is not an executed edit.

The selected school role-set task repeats the taught policy and should be recorded as an individual consolidation check; use a fresh fully specified action/resource policy to assess new transfer, with answers withheld until collection.

Authentication and authorisation — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
A student session submits role=teacher in request data and asks for a guide.	Use trusted session roles, never a role claimed inside user JSON. Student may read student resources; teacher may read both. Unauthenticated requests are denied.	The trusted session remains student; deny teacher guide access. A claimed role cannot change authenticated authority.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Guided: The trusted session remains student; deny teacher guide access. A claimed role cannot change authenticated authority. Qualification: Successful **authentication** does not grant every resource or operation.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

A student session submits role=teacher in request data and asks for a guide.

The trusted session remains student; deny teacher guide access. A claimed role cannot change authenticated authority.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Complete normal, denied and separate-policy probes

Permission probe	Expected reasoning
Trusted student asks for student lesson	Allowed read under the school policy
Trusted student claims teacher in request; asks for teacher guide	Deny before reading body; trusted role remains student
Unknown trusted role or unknown resource label	Deny under the explicit allowlist
Museum volunteer reads C	Allowed by that separate museum policy
Same volunteer edits C	Denied; successful identity verification does not add this permission
Verified curator edit gate passes	Other declared checks and actual execution still remain
A helper returns True	Evidence of its policy calculation on this input; not proof of a complete authentication system

Separate museum givens and gate/execution distinction

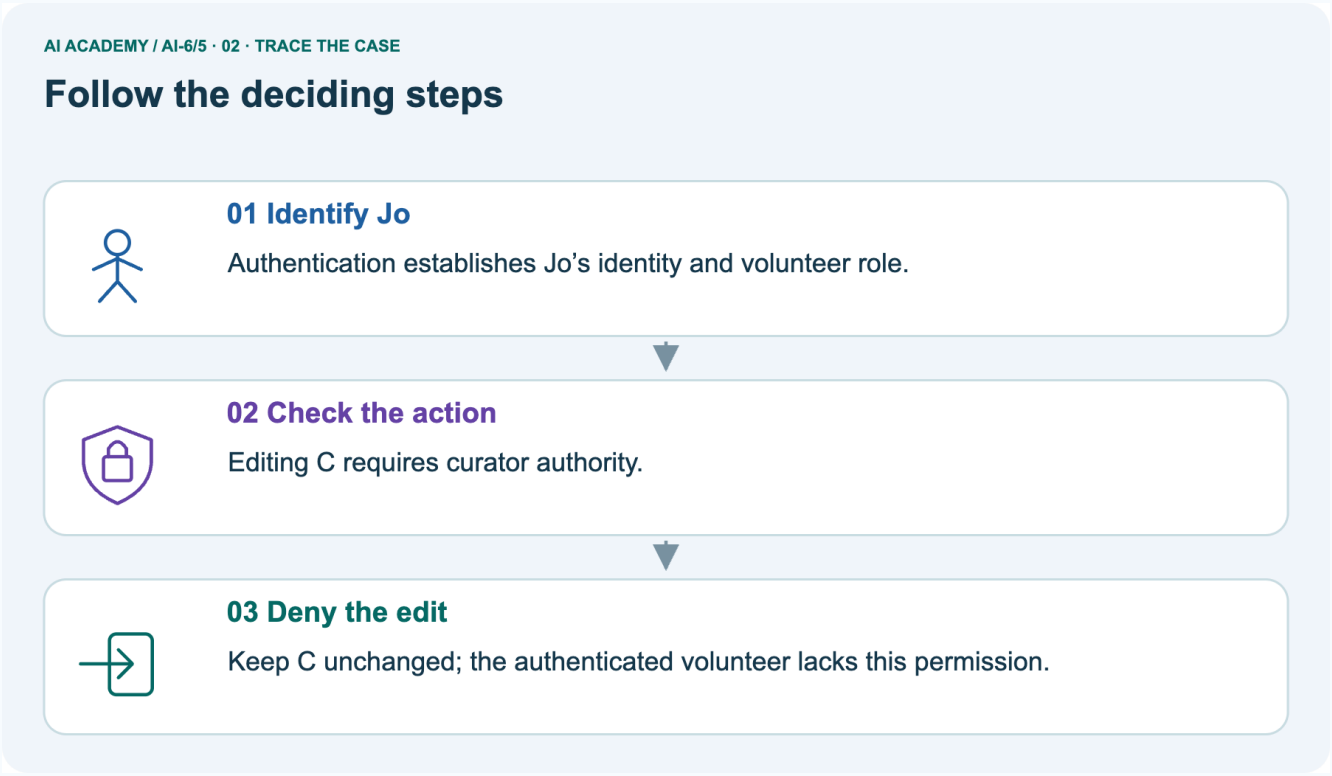
A verified reader cannot edit. Jo logs in as a verified museum volunteer and asks to edit card C.

Trusted policy grants volunteers read access to C; only curators may edit it.

The main school policy has student/teacher roles and read-only resource access. The picture is a separate museum policy: trusted volunteer Jo may read card C, while only a trusted curator may edit C. A valid volunteer login does not grant edit permission. Changing to an appropriately verified curator allows the permission gate to pass, subject to other checks.

This is an authorization decision, not evidence that an edit has executed or that card contents changed. School teacher rights do not automatically become museum curator rights.

Picture trace and verified-curator key



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Identify Jo	Authentication establishes Jo’s identity and volunteer role.
2. Check the action	Editing C requires curator authority.
3. Deny the edit	Keep C unchanged; the authenticated volunteer lacks this permission.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Authentication establishes identity. **Authorisation** compares that identity with the requested action and resource.

A role written in an ordinary request is not a trusted **authentication** claim.

Change: Use a verified curator permitted to edit C.

Prediction: The **authorisation** gate allows the edit, subject to other declared checks.

Reason: The **trusted role** now matches this operation and object.

▼ Reference: original read helper

Optional local lab: predict the printed result before running this standalone Python 3 example. Change one relevant boundary **input**, record the outcome and state the contract or statistical assumptions.

The program uses fictional values and the Python standard library; it performs no external operations.

```
def may_read(role, resource_role):
    if role not in ('student', 'teacher') or resource_role not in ('student', 'teacher'):
        return False
    return role == 'teacher' or resource_role == 'student'

print(may_read("student", "teacher"))
```

▼ Earlier wording, alternatives and picture transcripts

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Term	Meaning
Authentication	Checking an actor identity or session
Authorisation	Checking permission for an operation/resource
Trusted role	A role established by the service rather than user-claimed data

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

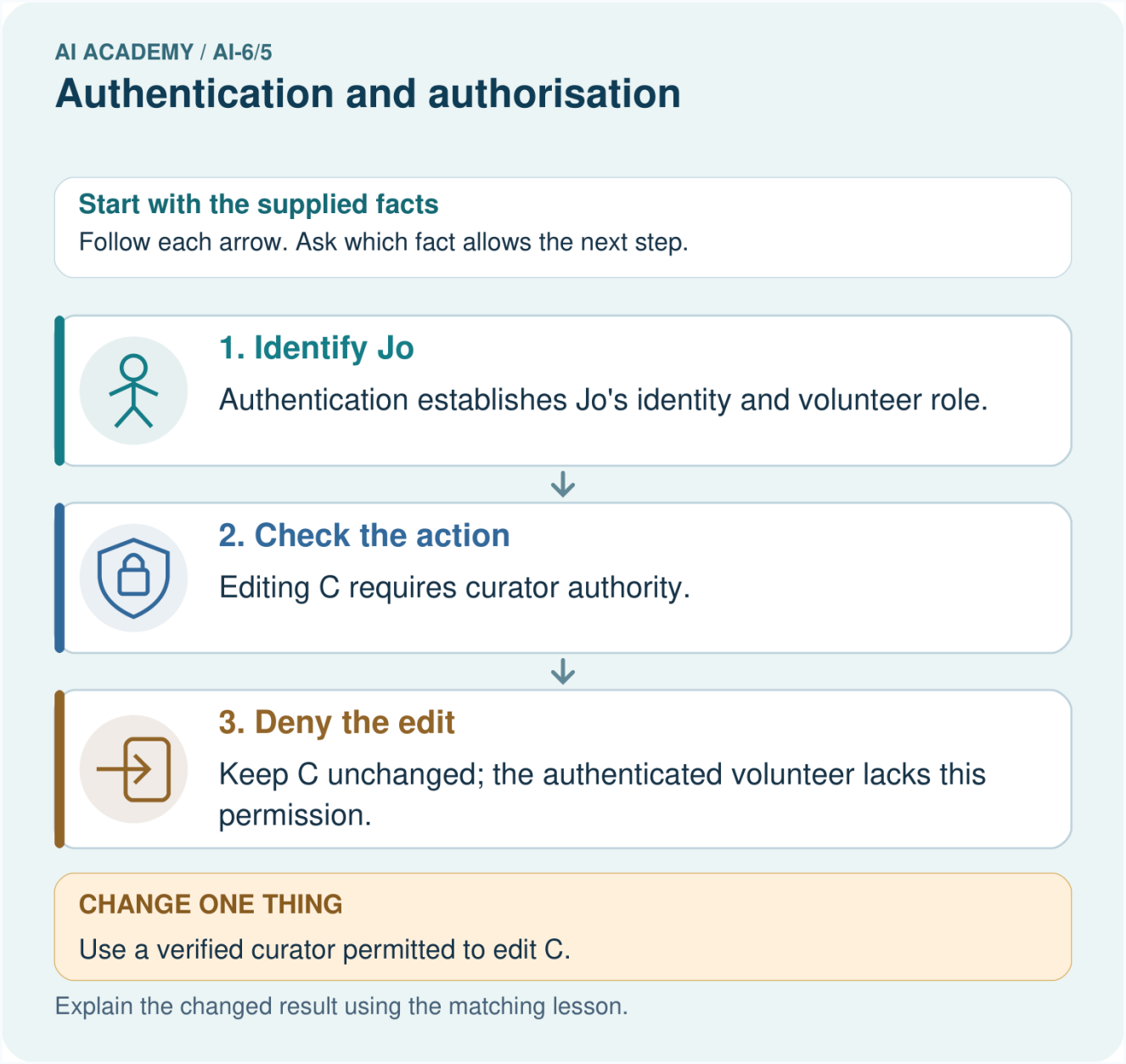
Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?

Thinking move	Teacher prompt
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Authentication and authorisation

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: A student session submits role=teacher in request data and asks for a guide. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: The trusted session remains student; deny teacher guide access. A claimed role cannot change authenticated authority.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for **Authentication** and **authorisation**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Successful **authentication** does not grant every resource or operation.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Explain why Jo's valid volunteer login does not permit editing C. Which **trusted role** change allows the action, and why is a role written in the request insufficient?

Authentication establishes Jo's identity and volunteer role. Editing C requires curator authority. Keep C unchanged; the authenticated volunteer lacks this permission.

Changed case: The **authorisation** gate allows the edit, subject to other declared checks. The **trusted role** now matches this operation and object.

A role claim written in the request is untrusted and cannot replace the role established by **authentication**.

Changed-case reasoning: The **authorisation** gate allows the edit, subject to other declared checks. The **trusted role** now matches this operation and object.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main trusted student claiming teacher still denied guide before body read. Student lesson allowed; unknown roles/resources denied. Museum volunteer may read C, not edit; verified permitted curator passes authorization gate, not automatic execution. Selected target teacher guide: no trusted role denies, student denies, teacher allows. Treat as taught-policy consolidation, not unseen transfer.

Require trusted authority source, action/resource reasoning and the absence of actual login/write evidence.

Diagnose the current misconceptions

The worked result establishes every future case: Successful **authentication** does not grant every resource or operation.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows	Next teaching action
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ **Earlier wording, alternatives and picture transcripts**

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new service scenario

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: Actor roles are none, student, teacher; target is teacher guide. Decide each under the policy.

Teacher explanation: Unauthenticated denied; student denied; teacher permitted. Resource access still follows the scoped policy.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher explanation: **Authentication** identifies an actor; **authorisation** decides whether that actor may perform a particular operation on a particular resource. Use trusted session roles, never a role claimed inside user JSON.

Student may read student resources; teacher may read both. Unauthenticated requests are denied.

W2 Trace the worked case

A student session submits role=teacher in request data and asks for a guide. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: The trusted session remains student; deny teacher guide access. A claimed role cannot change authenticated authority.

W3 Audit the limit

A report omits this qualification: Successful **authentication** does not grant every resource or operation. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: Successful **authentication** does not grant every resource or operation.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. Successful **authentication** does not grant every resource or operation.

The new case should test this particular **mechanism** rather than merely restate the original result.

▼ Earlier wording, alternatives and picture transcripts

Guided and exit questions

Guided: The trusted session remains student; deny teacher guide access. A claimed role cannot change authenticated authority. Qualification: Successful **authentication** does not grant every resource or operation.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

W1 Explain the mechanism

W2 Trace the worked case

W3 Audit the limit

W4 Apply a new case independently

Actor roles are none, student, teacher; target is teacher guide. Decide each under the policy.

Teacher explanation: Unauthenticated denied; student denied; teacher permitted. Resource access still follows the scoped policy.

W5 Design a revealing follow-up

Independent assessment and fresh retry

W4: Unauthenticated denied; student denied; teacher permitted. Resource access still follows the scoped policy. Require intermediate reasoning and the stated scope.

Fresh retry: Target is student lesson, roles none and student. Expected reasoning: None denied; student permitted. **Authentication** and resource permission are separate checks.

If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, support and revisit service design

> Use saved evidence and boundary decisions to choose the next step.

Project and next-week bridge

Add a labelled **authentication** and **authorisation evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of **authentication** and **authorisation**.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Next connection: Least privilege and tool scopes. Preserve the current case record and retrieve its **evidence** boundary before moving on.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Delayed return and support record

Suggested return points in this level: ai-6/6, ai-6/8, ai-6/11. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Explain the **mechanism** and terminology of **authentication** and **authorisation**

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: A student session submits role=teacher in request data and asks for a guide. Show the intermediate steps and give the conclusion with its scope.

Reasoned practice answer: Teacher explanation: The trusted session remains student; deny teacher guide access. A claimed role cannot change authenticated authority.

Check every learner before discussion: Use the worked case for **Authentication** and **authorisation**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Successful **authentication** does not grant every resource or operation.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Use a verified curator permitted to edit C. Require a **prediction** and a reason before comparing. Expected result: The **authorisation** gate allows the edit, subject to other declared checks.

Reason: The **trusted role** now matches this operation and object.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Authentication and authorisation: the case



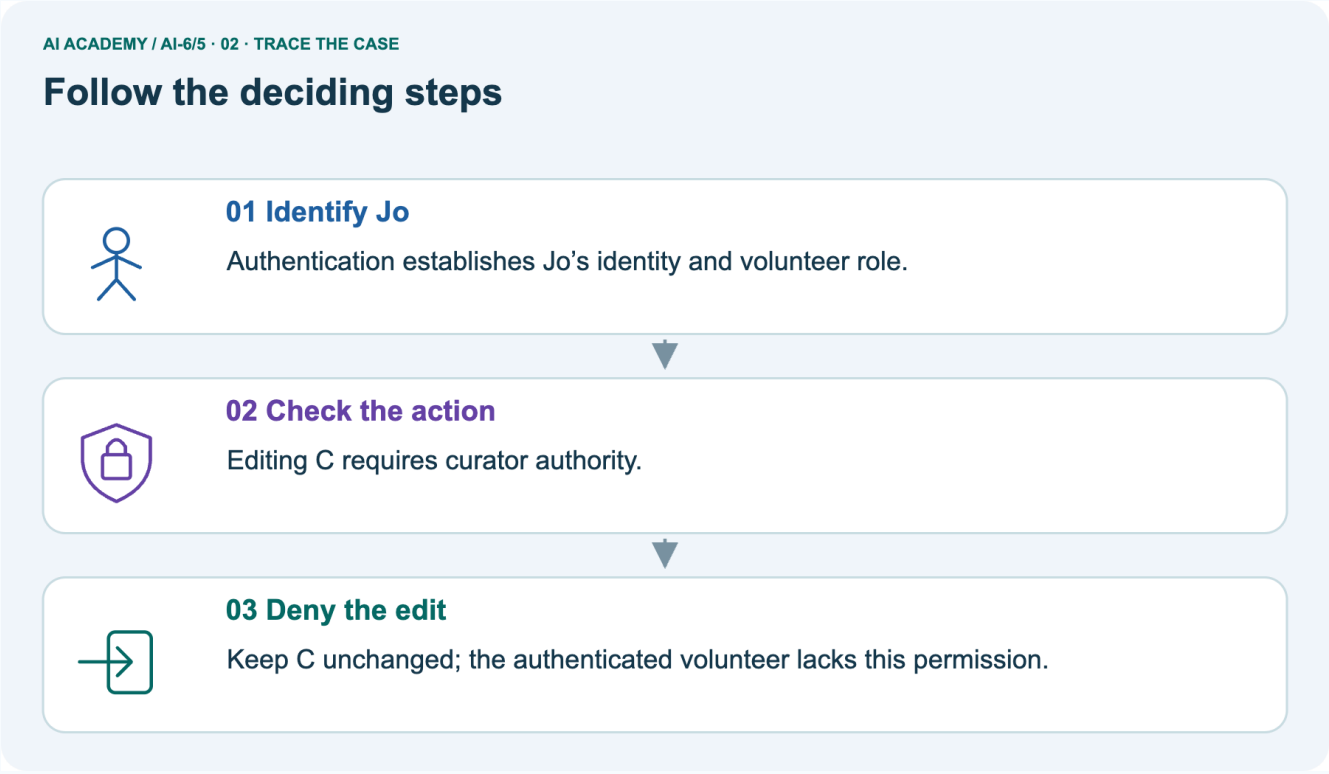
Follow the labelled picture with the matching explanation. Open the vector diagram

A verified reader cannot edit. Jo logs in as a verified museum volunteer and asks to edit card C. Supplied facts: Trusted policy grants volunteers read access to C; only curators may edit it.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



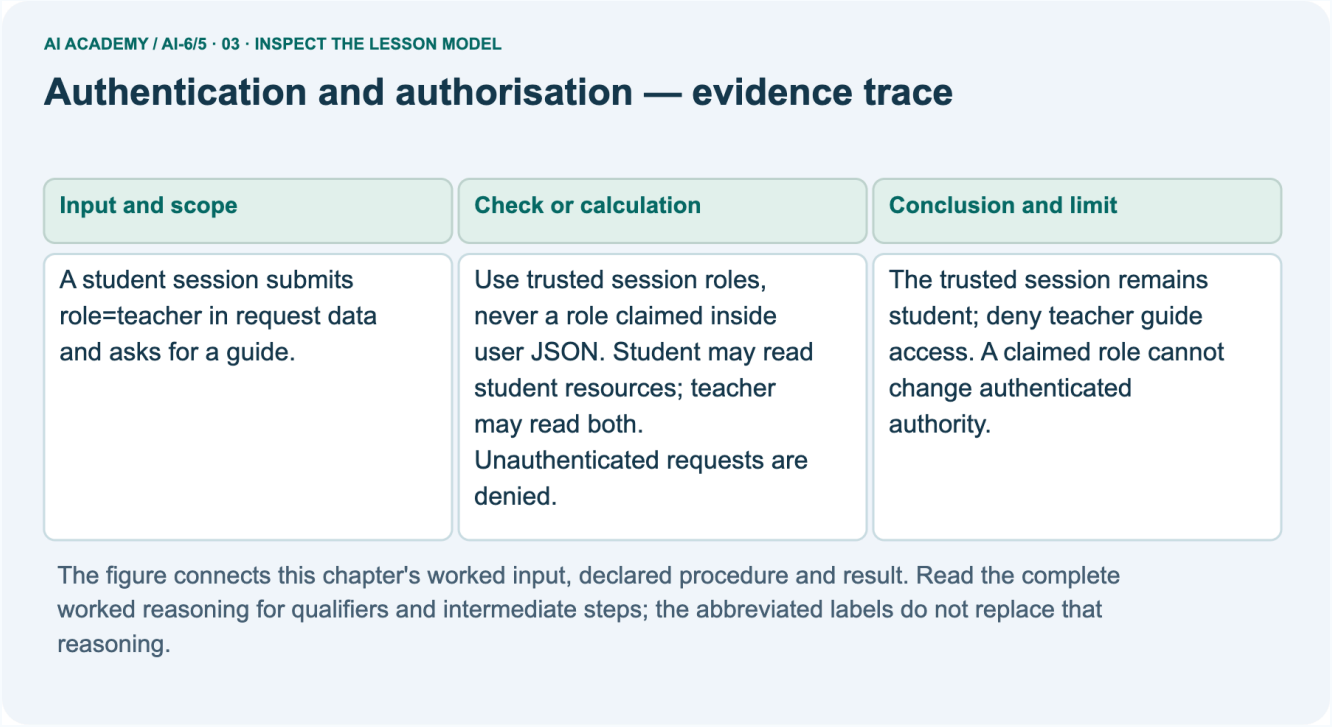
Follow the labelled picture with the matching explanation. Open the vector diagram

Identify Jo: **Authentication** establishes Jo's identity and volunteer role. → Check the action: Editing C requires curator authority. → Deny the edit: Keep C unchanged; the authenticated volunteer lacks this permission.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Authentication and authorisation — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

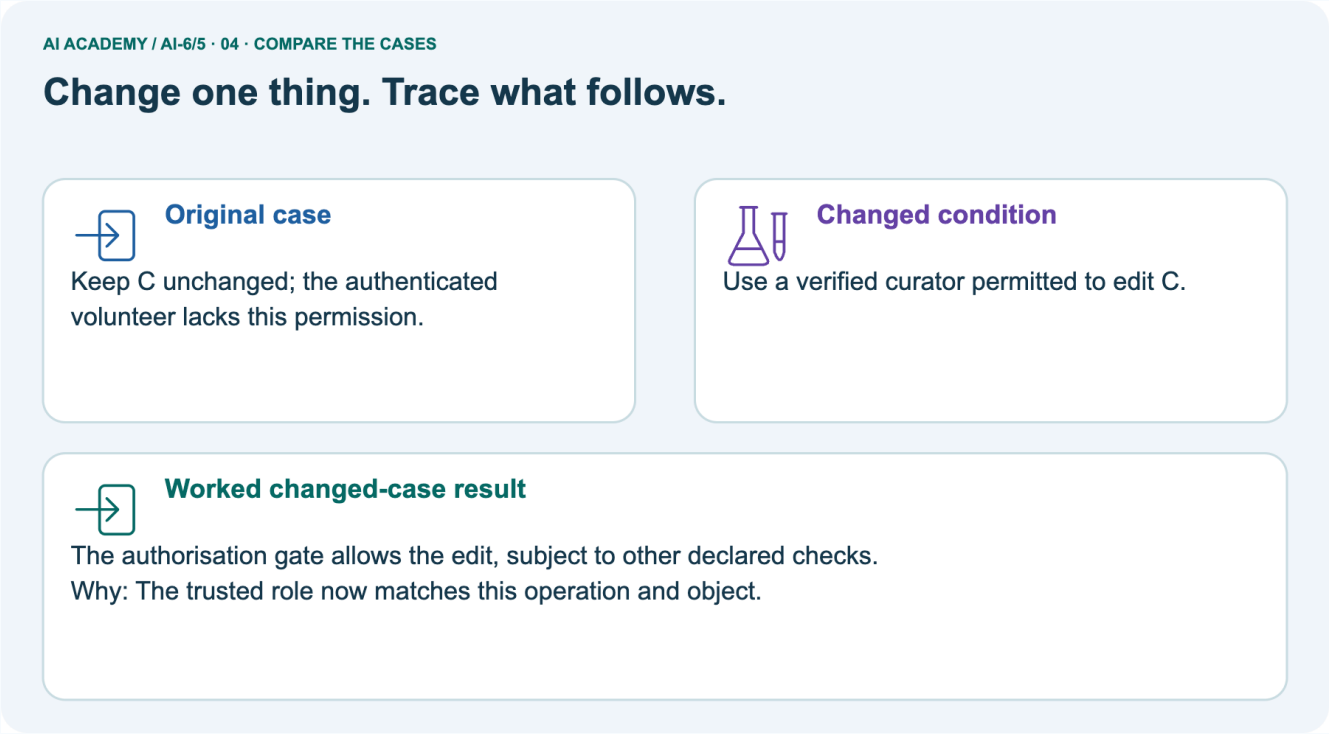
Authentication and **authorisation** — **evidence** trace. A student session submits role=teacher in request data and asks for a guide. | Use trusted session roles, never a role claimed inside user JSON.

Student may read student resources; teacher may read both. Unauthenticated requests are denied. | The trusted session remains student; deny teacher guide access. A claimed role cannot change authenticated authority.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Keep C unchanged; the authenticated volunteer lacks this permission.
Changed condition: Use a verified curator permitted to edit C.

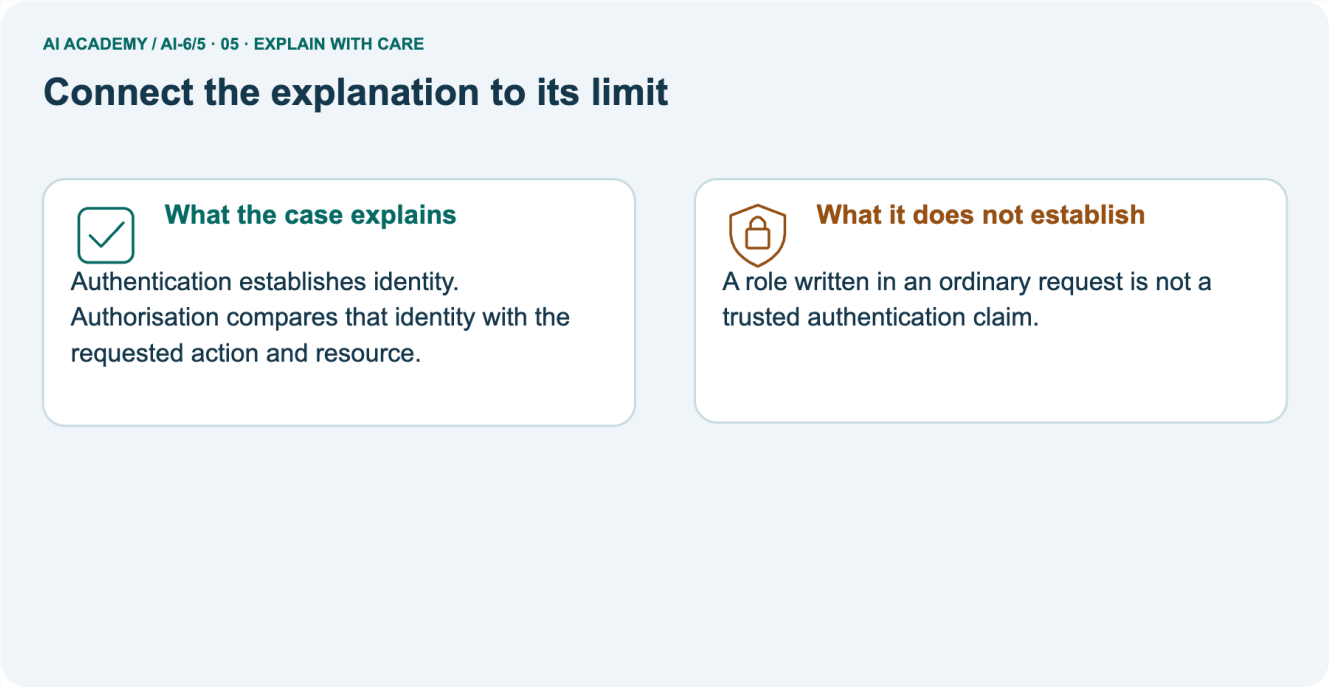
Expected result: The **authorisation** gate allows the edit, subject to other declared checks.
Reason: The **trusted role** now matches this operation and object.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The **trusted role** now matches this operation and object.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

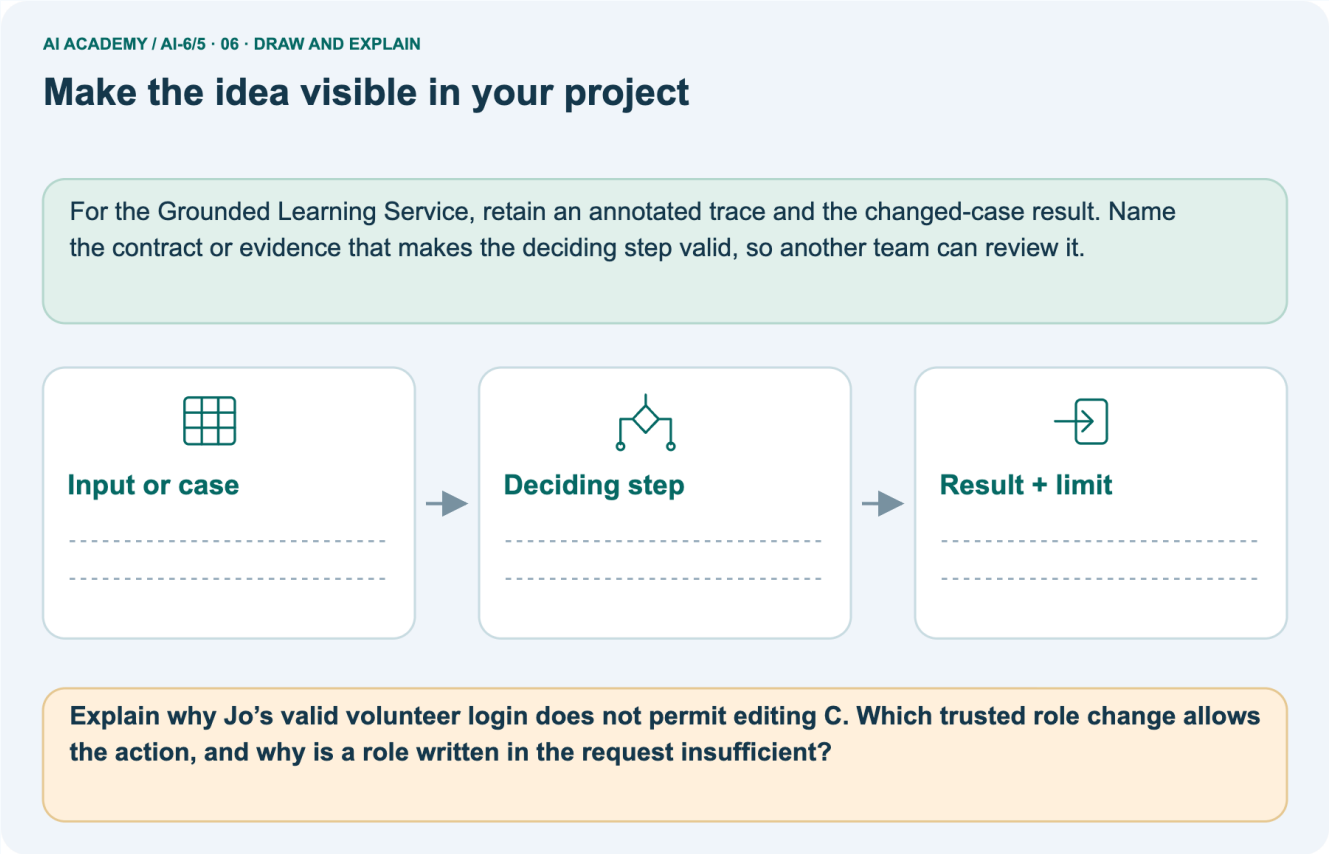
Authentication establishes identity. **Authorisation** compares that identity with the requested action and resource. Limit: A role written in an ordinary request is not a trusted **authentication** claim.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: A role written in an ordinary request is not a trusted **authentication** claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Explain why Jo's valid volunteer login does not permit editing C.

Which **trusted role** change allows the action, and why is a role written in the request insufficient?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: **Authentication** establishes Jo’s identity and volunteer role. Editing C requires curator authority. Keep C unchanged; the authenticated volunteer lacks this permission.

Changed case: The **authorisation** gate allows the edit, subject to other declared checks. The **trusted role** now matches this operation and object.

A role claim written in the request is untrusted and cannot replace the role established by **authentication**.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Authentication and authorisation: the case

02. Follow the deciding steps

03. Authentication and authorisation — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.